



JERUSALEM COLLEGE OF ENGINEERING

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Approved by AICTE, New Delhi, Affiliated to Anna University Chennai,
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Velachery Main Road, Narayanapuram, Pallikaranai, Chennai, Tamil Nadu - 600100.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

QUESTION BANK

Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No.	:	I
Unit Title	:	STORAGE SYSTEMS
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

Q. No.	Questions	Marks	BTL	CO
1.	What is digital data? Mention its types. Digital data is information represented and processed in binary form (0s and 1s) by computers and digital devices. It can include text, numbers, images, audio, and video. Digital data can be easily stored, processed, transmitted, and duplicated. It may be classified as structured, semi-structured, and unstructured data.	2	K1	CO1
2.	What are the different types of digital data? Digital data can be broadly classified into: Structured data – Organized in a fixed format, such as rows and columns in a relational database. Semi-structured data – Does not follow a rigid table structure but contains tags or markers, such as XML and JSON. Unstructured data – Does not have a predefined structure, such as images, videos, audio files, and social media posts.	2	K1	CO1
3.	What is information storage? Information storage is the process of saving and maintaining digital information so that it can be accessed when required. Data can be stored using HDDs, SSDs, magnetic tapes, and storage arrays. Storage provides persistent retention of data even when the system is powered off.	2	K1	CO1
4.	What are the key characteristics of a data center? A data center is a facility used to house computing and storage infrastructure. Availability – Provides continuous access to applications and data. Scalability – Resources can be increased or decreased according to requirements. Security and reliability – Protect infrastructure and data while ensuring dependable operation.	2	K1	CO1



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	Explain the evolution of computing platforms. The evolution of computing platforms represents the transition from centralized computing to highly distributed and connected environments. Mainframe era – Centralized computers served many users through terminals. Client-server and web computing – Processing became distributed between clients and servers. Cloud, mobile, and distributed computing – Computing resources became available through networks, mobile devices, and cloud platforms.	2	K2	CO1
6.	What is Information Lifecycle Management (ILM)? Information Lifecycle Management (ILM) is a systematic approach for managing information throughout its entire lifecycle. It manages information from creation to deletion. Data is stored according to its business value, access requirements, and importance.	2	K1	CO1
7.	List the major stages of the information lifecycle. The major stages of the information lifecycle are: Creation/Acquisition – Data is generated or collected. Use, Storage, and Management – Data is actively accessed, processed, and stored. Archiving/Disposal – Inactive data is archived for long-term retention or securely deleted when no longer required.	2	K1	CO1
8.	What are the benefits of Information Lifecycle Management? ILM provides several benefits to an organization Reduces storage cost by moving less important data to economical storage. Improves data availability by keeping frequently accessed data on appropriate storage. Supports compliance and security by ensuring proper retention and disposal of information.	2	K2	CO1
9.	What is cloud computing? Cloud computing is a technology that provides computing resources such as servers, storage, applications, and networks through a network on demand. Resources can be accessed on demand without owning the physical infrastructure.	2	K1	CO1
10.	What are the essential characteristics of cloud computing? According to the standard cloud computing model, there are five essential characteristics: On-demand self-service – Users can provision resources automatically. Broad network access and resource pooling – Resources are available over networks and shared among multiple users. Rapid elasticity and measured service – Resources can scale quickly and usage can be monitored and measured.	2	K1	CO1
11.	What are cloud service models? Cloud service models define the level of computing resources provided to users. IaaS (Infrastructure as a Service) – Provides virtual machines, storage, and networking infrastructure. PaaS (Platform as a Service) – Provides a platform for developing and deploying applications. SaaS (Software as a Service) – Provides ready-to-use software applications through the Internet.	2	K1	CO1



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	What are the cloud deployment models? Cloud deployment models describe how cloud infrastructure is owned, managed, and accessed. Public Cloud – Resources are provided by a cloud provider and shared among multiple customers. Private Cloud – Infrastructure is dedicated to a single organization. Hybrid Cloud – Combines private and public cloud environments to share workloads and resources.	2	K1	CO1
13.	What is Big Data Analytics? Big Data Analytics is the process of analyzing large, complex, and rapidly generated datasets to obtain useful information. It processes data having characteristics such as high volume, velocity, and variety. It identifies patterns, trends, and relationships within large datasets.	2	K1	CO1
14.	What is social networking? Mention its importance. Social networking refers to online platforms that allow users and organizations to communicate, interact, and share information. It enables real-time communication and collaboration. It allows users to share text, images, videos, and other information.	2	K2	CO1
15.	What is mobile computing? Mobile computing is a technology that allows users to access computing resources and services while moving from one location to another. It uses mobile devices such as smartphones, tablets, and laptops. It relies on wireless communication technologies such as Wi-Fi and cellular networks.	2	K1	CO1
16.	What are Third Platform Technologies? Third Platform Technologies represent a modern computing environment based on four major technologies: Cloud computing – Provides scalable computing resources and services. Big Data and Analytics – Processes large datasets to generate useful insights. Social networking and Mobile computing – Enables connected users, mobile access, communication, and collaboration.	2	K2	CO1
17.	What are the characteristics of Third Platform infrastructure? Third Platform infrastructure has the following characteristics: Scalable and flexible – Resources can be dynamically increased or decreased. Highly connected – Supports mobile devices, social networks, and cloud-based services. Data-driven and intelligent – Uses Big Data, analytics, and automation for decision-making.	2	K2	CO1
18.	What are the imperatives for Third Platform transformation? Third Platform transformation requires organizations to modernize their IT infrastructure and business processes. Cloud adoption – Organizations need flexible and scalable cloud-based infrastructure. Data-driven decision-making – Big Data analytics must be used to generate business insights.	2	K2	CO1
19.	What are the major building blocks of a data center? The major building blocks of a data center include: Compute systems – Servers that execute applications and process data. Storage systems – Store and provide access to organizational data. Network infrastructure – Connects servers, storage, users, and other	2	K1	CO1



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	devices.			
20.	What is compute virtualization and Software-Defined Data Center (SDDC)? Compute virtualization is the technique of creating multiple virtual computing environments on a single physical server. A hypervisor creates and manages multiple virtual machines. Software-Defined Data Center (SDDC) extends virtualization to the complete data center. Compute, storage, and networking resources are virtualized. Resources are managed and controlled through software and automation.	2	K2	CO1

PART – B

Q. No.	Questions	Marks	BT L	CO
1.	Explain the different types of digital data and information storage systems in detail. Discuss the key characteristics of a modern data center and explain the evolution of computing platforms from mainframe systems to modern cloud-based computing.	16	K2	CO 1
2.	Explain Information Lifecycle Management (ILM) in detail. Discuss the different stages of the information lifecycle, the importance of ILM, data classification, storage management, archiving, retention, and secure disposal of information.	16	K2	CO 1
3.	Explain cloud computing in detail. Discuss its essential characteristics, major cloud service models (IaaS, PaaS, SaaS), and cloud deployment models (Public, Private, Hybrid, and Community) with suitable examples.	16	K2	CO 1
4.	Discuss Third Platform Technologies in detail. Explain the role of cloud computing, Big Data analytics, social networking, and mobile computing. Also explain the major characteristics of Third Platform infrastructure and the imperatives for Third Platform transformation.	16	K2	CO 1
5.	A large organization is experiencing rapid growth in data, increasing mobile users, and fluctuating application workloads. Analyze how Third Platform Technologies can be applied to address these challenges. Explain the roles of cloud computing, Big Data analytics, social networking, and mobile computing in the proposed solution.	16	K3	CO 1
6.	Explain the major building blocks of a modern data center in detail. Discuss compute systems, storage systems, network infrastructure, power and cooling systems, security, and data center management. Explain how these components work together to provide reliable IT services.	16	K2	CO 1
7.	A multinational organization is planning to modernize its existing data center to support increasing data volumes, unpredictable workloads, remote users, and business-critical applications. Apply the concepts of	16	K3	CO 1



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	cloud computing, cloud deployment models, compute virtualization, and Software-Defined Data Center (SDDC) to propose a suitable data center environment. Explain how each technology can improve scalability, resource utilization, flexibility, availability, and cost efficiency.			
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QUESTION BANK

Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No	:	II
Unit Title	:	INTELLIGENT STORAGE SYSTEMS AND RAID
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

Q.No.	Questions	Marks	BTL	CO
1.	What is an intelligent storage system? It is a storage system with built-in intelligence for managing data and storage resources. It provides features such as caching, RAID, replication, and automated management.	2	K1	CO2
2.	List any two major components of an intelligent storage system. Storage processors/controllers. Cache memory and storage devices such as HDDs or SSDs.	2	K1	CO2
3.	What is the function of a storage controller? It manages communication between servers and storage devices. It controls data access, RAID operations, and storage management functions.	2	K2	CO2
4.	What is storage cache? Storage cache is high-speed memory used to temporarily store frequently accessed data. It improves storage performance by reducing data access latency.	2	K2	CO2
5.	What is meant by addressing in a storage system? Addressing identifies the location of data blocks on a storage device. It enables the system to locate and access the required data efficiently.	2	K1	CO2
6.	What is LBA in hard disk drives? LBA stands for Logical Block Addressing . It identifies each logical block using a unique numerical address.	2	K1	CO2



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	Name any two important performance parameters of an HDD. Seek time – time required to position the read/write head. Rotational latency – time required for the desired sector to rotate under the read/write head.	2	K1	CO2
8.	Differentiate HDD and SSD based on their storage mechanism. HDD stores data magnetically on rotating platters. SSD stores data electronically using NAND flash memory.	2	K2	CO2
9.	What is an SSD? SSD stands for Solid-State Drive and uses flash memory for data storage. It has no mechanical moving parts and provides faster data access.	2	K1	CO2
10.	Why are SSDs generally faster than HDDs? SSDs do not require mechanical head movement or platter rotation. They provide lower access latency and faster I/O operations.	2	K2	CO2
11.	What is RAID? RAID stands for Redundant Array of Independent Disks . It combines multiple disks to improve performance, reliability, or both.	2	K1	CO2
12.	What is the main purpose of RAID 0? RAID 0 uses striping to distribute data across multiple disks. It improves read/write performance but provides no fault tolerance.	2	K2	CO2
13.	What is the main advantage of RAID 1? RAID 1 uses mirroring , storing identical data on two disks. It provides data protection against the failure of one disk.	2	K2	CO2
14.	What is RAID 5? RAID 5 uses block-level striping with distributed parity. It can tolerate the failure of one disk without losing data.	2	K1	CO2
15.	What is RAID 6? RAID 6 uses distributed dual parity. It can tolerate the failure of two disks simultaneously.	2	K1	CO2
16.	What is meant by scale-up storage architecture? Scale-up increases the capacity and performance of an existing storage system. It can be achieved by adding disks, controllers, cache, or other resources.	2	K2	CO2
17.	What is meant by scale-out storage architecture? Scale-out increases storage capacity and performance by adding storage nodes. Multiple nodes work together as a unified storage system.	2	K2	CO2
18.	Differentiate scale-up and scale-out storage. Scale-up: expands resources within an existing storage system. Scale-out: expands the system by adding additional storage nodes.	2	K2	CO2
19.		2	K1	CO2



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	Name the two common approaches to storage system scaling. Scale-up architecture – expansion within an existing storage system. Scale-out architecture – expansion by adding independent storage nodes.			
20.	A storage system requires high performance but does not require data redundancy. Which RAID level is suitable and why? RAID 0 is suitable because it distributes data across multiple disks using striping. It provides high read/write performance but does not provide fault tolerance.	2	K3	CO2

PART – B

Q. No.	Questions	Marks	BT L	CO
1.	Explain the components of an intelligent storage system and describe the functions of storage processors, cache, connectivity, and storage devices.	16	K2	CO 2
2.	Explain the components, addressing mechanisms, and performance characteristics of Hard Disk Drives (HDDs) and Solid-State Drives (SSDs). Compare HDD and SSD based on their architecture and performance.	16	K3	CO 2
3.	Explain RAID technology in detail. Discuss RAID 0, RAID 1, RAID 3, RAID 5, and RAID 6 with their architecture, working principles, advantages, and limitations.	16	K2	CO 2
4.	Analyze the different RAID levels based on data protection, performance, storage efficiency, and fault tolerance. Recommend a suitable RAID level for a high-availability enterprise application with justification.	16	K4	CO 2
5.	Explain the different types of intelligent storage systems and discuss their major features, benefits, and applications in enterprise environments.	16	K2	CO 2
6.	A large organization requires a storage infrastructure that can support increasing data volumes, high performance, and continuous availability. Analyze the requirements and propose a suitable intelligent storage architecture using RAID, HDD/SSD, and Scale-Up or Scale-Out technologies. Justify your design.	16	K4	CO 2

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Regulation	:	2023
Course Name	:	STORAGE TECHNOLOGIES
Course Code	:	JAL2007
Year / Semester / Branch	:	III / V / AI & ML
Unit No	:	III
Unit Title	:	STORAGE NETWORKING TECHNOLOGIES AND VIRTUALIZATION
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

Q. No.	Questions	Marks	BTL	CO
1.	What is a Block-Based Storage System? It stores data as fixed-size blocks that are independently addressable. It is commonly used in SAN environments and databases.	2	K1	CO3
2.	What is a File-Based Storage System? It stores data as files organized into directories and folders. Users and applications access data through a file system.	2	K1	CO3
3.	What is Object-Based Storage? It stores data as objects containing data, metadata, and a unique identifier. It is highly suitable for large-scale and unstructured data storage.	2	K1	CO3
4.	What is Unified Storage? Unified storage supports both file-based and block-based storage from a single system. It provides flexible storage management for different applications.	2	K2	CO3
5.	What is Fibre Channel SAN? Fibre Channel SAN is a high-speed dedicated network used to connect servers with storage systems. It provides reliable and low-latency block-level data access.	2	K1	CO3
6.	Name any two major components of an FC SAN. Host Bus Adapter (HBA) connects the server to the Fibre Channel network. FC Switch provides connectivity between servers and storage devices.	2	K1	CO3
7.	What is zoning in Fibre Channel SAN?	2	K2	CO3



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	Zoning controls which devices in an FC SAN can communicate with each other. It improves security and reduces unnecessary SAN traffic.			
8.	What is link aggregation in an FC SAN environment? Link aggregation combines multiple physical links to provide increased bandwidth. It can improve availability and provide redundancy between connected devices.	2	K2	CO3
9.	What is virtualization in an FC SAN environment? SAN virtualization abstracts physical storage resources and presents them as logical storage resources. It improves storage utilization, flexibility, and management.	2	K2	CO3
10.	What is the role of Software-Defined Networking (SDN) in storage networks? SDN separates the network control plane from the data-forwarding plane. It enables centralized and programmable management of network resources.	2	K2	CO3

PART – B

Q. No.	Questions	Marks	BT L	CO
1.	Explain the architecture and major components of a Fibre Channel SAN. Describe the role of FC switches, host bus adapters, and storage devices.	8	K2	CO 3
2.	Explain virtualization in an FC SAN environment and discuss how Software-Defined Networking (SDN) can improve the flexibility, management, and scalability of storage networks.	8	K4	CO 3
3.	Compare Block-Based, File-Based, and Object-Based Storage systems based on their architecture, data access method, scalability, and typical applications.	8	K3	CO 3
4.	Explain FC SAN topologies, link aggregation, and zoning. Analyze how these mechanisms improve connectivity, performance, availability, and security in a Fibre Channel SAN.	8	K4	CO 3

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